

Development - phase 2.1

Thursday, 28 May 2026

8:06 PM

Current Design Review

GOOD THINGS

You already separated:

prototype-docs
from:

science-project generation

This is correct.

You are also tracking:

- photos
- notes
- cost sheets
- records

which is VERY important later.

Suggested Improvements

☐ **Rename Main Doma**_{in}

Instead of:

prototype-docs

Better:

prototype-management

OR

prototype-execution

Reason:

Later it will include:

- workflows
- build status
- prototype lifecycle
- approvals
- validations

not just docs.

☐ **Improve Endpoint Hierarc**_{hy}

Current:

/api/v1/prototypes/{prototype_id}/photos

This is GOOD.

Keep this style.



Recommended Structure

/api/v1/prototypes/

/api/v1/prototypes/{id}

/api/v1/prototypes/{id}/photos

/api/v1/prototypes/{id}/notes

/api/v1/prototypes/{id}/cost-sheets

/api/v1/prototypes/{id}/build-records

/api/v1/prototypes/{id}/materials

/api/v1/prototypes/{id}/status



Small Naming Improvements

Instead of:

records

Better:

build-records

because:

“records” becomes ambiguous later



Cost Sheet Endpoint

Current:

POST /cost-sheet

Better:

POST /cost-sheets

GET /cost-sheets

Reason:

Later:

- multiple revisions
- material comparisons
- pricing experiments

may happen.



Add Prototype Status Early

Add Prototype Status Early

VERY important.

Add:

/status

Possible values:

- planned
- material_pending
- in_progress
- completed
- failed
- needs_revision

This becomes useful VERY quickly.



Add Materials Endpoint Early

Because later:

- material analytics
- cheapest sourcing
- reusable inventory

becomes valuable.

Add:

/materials



Future Important Insight

Eventually:

science-project

and:

prototype

are DIFFERENT entities.

Example

Science Project

Hydraulic Bridge

Prototype Builds

Could have:

- cardboard version
- wood version
- low-cost version
- festival demo version

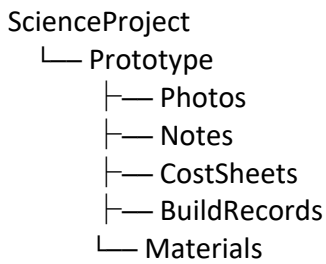
So:

one project can have multiple prototypes

VERY important future distinction.



Suggested Future Data Relationship



MOST IMPORTANT THING

You are currently building:

execution intelligence

not just:

AI text generation

That distinction is VERY powerful long-term.



FINAL VERDICT

Your direction:

✓ Correct

Your hierarchy:

✓ Mostly correct

Improvements recommended:

- rename records → build-records
- pluralize cost-sheets
- add status endpoint
- add materials endpoint
- think project vs prototype separately

That will make your architecture MUCH cleaner later.

what is the purpose can you explain not getting it

You are currently thinking:

AI generates project → done

But your REAL system is actually:

AI generates project



Human builds real prototype



System captures execution intelligence



Future generations improve

That second part is why these endpoints exist.



Let Me Explain With Real Example

Suppose AI generated:

Hydraulic Bridge Model

AI says:

- use syringes
- use cardboard
- use pipes
- estimated cost ₹250

Now your assistant actually tries building it.

What Happens In Real Life?

Assistant discovers:

- cardboard bends
- pipes leak
- cost became ₹480
- build took 2 days
- glue was insufficient
- children struggled assembling it

THIS is:

real execution intelligence

AI did not know this.

Your system must capture this knowledge.



WHY EACH ENDPOINT EXISTS



1 Photos Endpoi_{nt}

POST /photos

Purpose:

Store:

- final prototype images
- build progress images
- proof prototype was actually built

Why important?

Later:

- teachers see examples
- students understand better
- demo showcases become possible
- future AI can reference successful builds



2 Notes Endpoi_{nt}

POST /notes

Purpose:

Capture:

- build difficulties
- practical issues

- improvements
- observations

Example note:

Cardboard became unstable after water exposure

This becomes:

practical learning database

Build Records Endpoi_{nt}

GET /build-records

Purpose:

Track:

- who built it
- when built
- build duration
- build attempts
- success/failure

Why?

Later you may have:

- multiple assistants
- workshops
- students building projects

Need operational tracking.

Cost Sheet Endpoi_{nt}

POST /cost-sheets

Purpose:

Track:

- actual material costs
- travel costs
- build costs
- time investment

Why important?

Because later:

feasibility matters

Example:

AI says:

₹200 project

Reality:

₹950 project

Now system learns:

- unrealistic outputs
- expensive materials
- optimization opportunities

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Materials Endpoint (Suggested)

Purpose:

Store:

- actual materials used
- alternatives
- local market availability

Example:

Thermocol unavailable locally

→ assistant used foam board

VERY valuable later.



Status Endpoint (Suggested)

Purpose:

Track prototype lifecycle.

Example:

planned

→ in_progress

→ completed

→ failed

This becomes important when:

- many prototypes exist
- assistant works remotely
- you manage builds asynchronously



MOST IMPORTANT THING YOU ARE BUILDING

Not:

AI generated projects

You are building:

educational execution intelligence

VERY different.



Long-Term Strategic Value

Over time your system learns:

AI Assumption	Real World Learning
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cheap project	actually expensive
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easy build	children struggle
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safe material	breaks easily
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simple workflow	takes too much time
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This becomes:

your proprietary educational knowledge



What This Is Powerful

Why This is Powerful?

Most AI systems stop here:

text generation

Your system continues into:

real-world execution validation

That is MUCH harder to replicate later.

SIMPLE ANALOGY

Think like this:

AI Layer

recipe generator

Prototype Layer

actual cooking experience

Without actual cooking:

- recipe quality never improves.

Same here.